

USEPA DMR-QA 36

NPDES PERMITTEE DATA REPORT FORM



ERA

A Waters Company

Due August 26, 2016

USEPA NPDES MI0020257

Permit #:

Permit Ext:

Permittee Name: Cadillac WWTP

Facility Address: 1121 Plett Rd.

City: Cadillac

State: MI

Postal Code: 49601

Phone Number: (231) 775-2368

Fax Number: (231) 775-2876

E-mail address: lab@cadillac-mi.net

Optional: If WP study was used, list PT provider name:

Optional: WP study number(s):

For DMRQA-36, conducted in 2016, the Permittee ensured that their laboratory(s) performing the required analyses:

Received PT Samples

Yes ☒ No ☐

Submitted Complete and Accurate Data
by July 1, 2016

Yes ☒ No ☐

Received a Graded Report by July
29, 2016

Yes ☒ No ☐

Certification by Permit Holder or Authorized Representative

(as per 40 C.F.R. Section 122.22)

I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate and complete. Each reported value was produced from a single analytical run using the analytical system that routinely performs these analyses to produce compliance monitoring data required under our National Pollutant Discharge Elimination System (NPDES) Permit. Neither I nor any of my subordinates compared our results from independent analyses conducted by us or any other laboratory before we reported our results to the USEPA. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.

Name of
Certifying Official: Amy Vail

Title: Lab Manager

Signature:

Date Signed:

Mailing Address: 200 N Lake Street

(enter only if different
from address above)

City: Cadillac

State: MI

Postal Code: 49601

Phone Number: (231) 775-2368

E-mail address: lab@cadillac-mi.net

RECEIVED

AUG 02 2016

DEQ
WATER RESOURCES DIVISION
GRAND RAPIDS

Please make corrections in boxes above

United States
ENVIRONMENTAL PROTECTION AGENCY
 Laboratory Performance Evaluation
 Laboratory DMR-QA Evaluation Study 36

USEPA NPDES Permit #: MI0020257

Permit Ext:

Identification of all CHEM, MICRO and TOX laboratories who did analyses for this permit							
Name of Laboratory	Address of Laboratory	U.S. EPA Lab Code	Lab Analysis Check box(es) that apply			Lab Type*	State-certified Lab**
			CHEM	MICRO	TOX		
Cadillac WWTP	200 N. Lake St. Cadillac, MI 49601	MI00082	☒	☒	☐	G	☒
Paragon Laboratories, Inc.	12649 Richfield Ct Livonia, MI 48150	MI01187	☐	☐	☒	C	☒

See Next Sheet for Lab Info

* Lab Types: C = Commercial F = Federal G = Local Government I = Industrial O = Other S = State

** See Footnote 2 on DMRQA-36, Frequently Asked Questions page



United States Environmental Protection Agency
Office of Enforcement and Compliance Assurance
Washington, DC 20460

DMR-QA Study 36

(This data is collected under the authority of Section 308 of the Clean Water Act.)

Permittee Name

State

NPDES Permit No.

Permit Extension

Cadillac WWTP

MI

0010757

Identification of all CHEM, MICRO and WET laboratories who performed analyses for this permit

Name of Laboratory	Address of Laboratory	U.S. EPA Lab Code	Lab Analysis Check box(es) that apply			Lab Type*	State-certified Lab**
			Chem	Micro	WET		
Kar Laboratories	4425 Manchester Rd Kalamazoo, MI 49001	MI00041	☒	☐	☐	C	☒
			☐	☐	☐		☐
			☐	☐	☐		☐
			☐	☐	☐		☐
			☐	☐	☐		☐
			☐	☐	☐		☐

* Lab Types: C = Commercial F = Federal G = Local Government I = Industrial O = Other S = State

** See Footnote on page 2 (Frequently Asked Questions) for the current list of states with lab accreditation programs

If you need additional space, please make a copy of this page for additional laboratories.

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00082

Chemistry/Microbiology Analyte Checklist

DMRQA Study 36

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Minerals Alkalinity as CaCO ₃ Chloride Conductivity at 25°C Fluoride Potassium Sodium Sulfate Total Dissolved Solids at 180°C Total Solids at 105°C				
Hardness Total Suspended Solids SM 2540 D-2011 Calcium Magnesium Calcium Hardness as CaCO ₃ Total Hardness as CaCO ₃	☐	☒	☐	☐
pH pH SM 4500-H+ B-2011	☐	☒	☐	☐
Settleable Solids Settleable Solids				
Solids Concentrate Total Suspended Solids Total Dissolved Solids at 180°C Total Solids at 105°C				
Solids Total Suspended Solids Total Dissolved Solids at 180°C Total Solids at 105°C				
Simple Nutrients Ammonia as N SM 4500-NH ₃ D-2011 Nitrate + Nitrite as N Nitrate as N ortho-Phosphate as P	☐	☒	☐	☐
Complex Nutrients Total Kjeldahl Nitrogen Total phosphorus as P SM 4500-P E-2011	☐	☒	☐	☐
Nitrite Nitrite as N				

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00082

Chemistry/Microbiology Analyte Checklist

DMRQA Study 36

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Demand BOD CBOD SM 5210 B-2011 COD TOC	☐	☒	☐	☐
Oil & Grease Concentrate n-Hexane Extractable Material(O&G) (Grav)				
Oil & Grease n-Hexane Extractable Material(O&G) (Grav) n-Hexane Extractable Material(O&G) (IR)				
Trace Metals Aluminum Antimony Arsenic Barium Beryllium Boron Cadmium Chromium Cobalt Copper Iron Lead Manganese Molybdenum Nickel Selenium Silver Strontium Thallium Vanadium Zinc				
Low-Level Mercury Low Level Mercury				
Mercury Mercury				
Hexavalent Chromium Hexavalent Chromium				
Turbidity Turbidity				

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00082

Chemistry/Microbiology Analyte Checklist

DMRQA Study 36

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
<u>Total Cyanide</u>				
Cyanide, total				
Amenable Cyanide				
<u>Total Phenolics (4-AAP)</u>				
Phenolics, total				
<u>Total Residual Chlorine</u>				
Total Residual Chlorine				
<u>Low-Level Total Residual Chlorine</u>				
Low Level Total Residual Chlorine				
<u>WasteWatR™ Coliform MicrobE™</u>				
Total Coliforms (MF)				
Fecal Coliforms (MF)				
SM 9222 D (m-FC)-2006				
E.coli (MF)				
Total Coliforms (MPN)				
Fecal Coliforms (MPN)				
E.coli (MPN)				

Print Name

Signature/Title

Date

Use a separate checklist for EACH lab used

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI01187

WET Organisms/Test Conditions/End Points Checklist

DMRWet Study 36

Analyte Number	Organisms / Conditions	End Points	Test Req	Laboratory's Graded Result		Analyte determined by state-certified lab
				Acceptable	Not Acceptable (Corrective Action Required)	
Fathead minnow (Test Code 13)						
0754	48Hr., Acute, Non-Renewal, 25° C, MHSF EPA 2000.0	LC50	☐	☒	☐	☐
Fathead minnow (Test Code 14)						
0755	48Hr., Acute, Non-Renewal, 25° C, 20% DMW	LC50				
Fathead minnow (Test Code 15)						
0808	7-day Short term Chronic, Daily Renewal, MHSF EPA 1000	IC25 (ON) Growth	☐	☒	☐	☐
0810	7-day Short term Chronic, Daily Renewal, MHSF EPA 1000	NOEC (ON) Growth	☐	☒	☐	☐
0756	7-day Short term Chronic, Daily Renewal, MHSF EPA 1000	NOEC Survival	☐	☒	☐	☐
Fathead minnow (Test Code 16)						
0812	7-day Short term Chronic, Daily Renewal, 20% DMW	IC25 (ON) Growth				
0814	7-day Short term Chronic, Daily Renewal, 20% DMW	NOEC (ON) Growth				
0759	7-day Short term Chronic, Daily Renewal, 20% DMW	NOEC Survival				
Ceriodaphnia dubia (Test Code 19)						
0764	48Hr., Acute Renewal, 25° C, MHSF EPA 2002	LC50	☐	☒	☐	☐
Ceriodaphnia dubia (Test Code 20)						
0765	48Hr., Acute Renewal, 25° C, 20% DMW	LC50				
Ceriodaphnia dubia (Test Code 21)						
0767	7-day Short term Chronic, Daily Renewal, MHSF EPA 1002	IC25 Reproduction	☐	☒	☐	☐
0768	7-day Short term Chronic, Daily Renewal, MHSF EPA 1002	NOEC Reproduction	☐	☒	☐	☐
0766	7-day Short term Chronic, Daily Renewal, MHSF EPA 1002	NOEC Survival	☐	☒	☐	☐
Ceriodaphnia dubia (Test Code 22)						
0770	7-day Short term Chronic, Daily Renewal, 20% DMW	IC25 Reproduction				
0771	7-day Short term Chronic, Daily Renewal, 20% DMW	NOEC Reproduction				
0769	7-day Short term Chronic, Daily Renewal, 20% DMW	NOEC Survival				
Daphnia magna (Test Code 32)						
0788	48Hr., Acute, Non-Renewal, 25° C, MHSF	LC50				
Daphnia pulex (Test Code 38)						
0794	48Hr., Acute, Non-Renewal, 25° C, MHSF	LC50				

Permittee Name: Cadillac WWTP

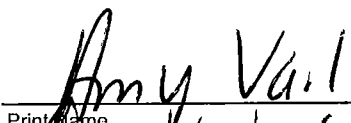
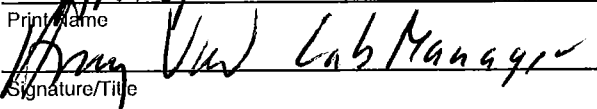
Permit Number: MI0020257

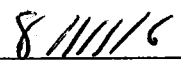
EPA Lab Code: MI01187

WET Organisms/Test Conditions/End Points Checklist

DMRWet Study 36

Analyte Number	Organisms / Conditions	End Points	Test Req	Laboratory's Graded Result		Analyte determined by state-certified lab
				Acceptable	Not Acceptable (Corrective Action Required)	
<u>Mysid (Test Code 42)</u>						
0798	48Hr., Acute, Non-Renewal, 25° C, 40 FSW	LC50				
<u>Mysid (Test Code 43)</u>						
0816	7-day Short term Chronic, Daily Renewal, 40 FSW	IC25 (ON) Growth NOEC (ON) Growth NOEC Survival				
0818	7-day Short term Chronic, Daily Renewal, 40 FSW					
0799	7-day Short term Chronic, Daily Renewal, 40 FSW					
<u>Inland silverside (Test Code 44)</u>						
0803	48Hr., Acute, Non-Renewal, 25° C, 40 FSW	LC50				
<u>Inland Silverside (Test Code 45)</u>						
0825	7-day Short term Chronic, Daily Renewal, 40 FSW	IC25 (ON) Growth NOEC (ON) Growth NOEC Survival				
0826	7-day Short term Chronic, Daily Renewal, 40 FSW					
0824	7-day Short term Chronic, Daily Renewal, 40 FSW					
<u>Sheepshead minnow (Test Code 46)</u>						
0804	48Hr., Acute, Non-Renewal, 25° C, 40 FSW	LC50				
<u>Sheepshead minnow (Test Code 47)</u>						
0820	7-day Short term Chronic, Daily Renewal, 40 FSW	IC25 (ON) Growth NOEC (ON) Growth NOEC Survival				
0822	7-day Short term Chronic, Daily Renewal, 40 FSW					
0805	7-day Short term Chronic, Daily Renewal, 40 FSW					


 Print Name

 Signature/Title Lab Manager


 Date

Use a separate checklist for EACH lab used

Permittee name Cadillac Water State MI NPDES Permit No. 0000057 EPA Lab Code MI00041

Chemistry/Microbiology Analyte Checklist

DMR-QA Study 36

Analyte Test	Test Required	Method Number Used (optional)	Laboratory's Graded Result		Analyte determined by state-certified lab*
			Acceptable	Not Acceptable (Corrective Action Required)	
Microbiology					
E. coli, MF or MPN	☐		☐	☐	☐
Fecal Coliform, MF or MPN	☐		☐	☐	☐
Total Coliform, MF or MPN	☐		☐	☐	☐
Trace Metals					
Aluminum	☐		☐	☐	☐
Antimony	☐		☐	☐	☐
Arsenic	☐		☐	☐	☐
Barium	☐		☐	☐	☐
Beryllium	☐		☐	☐	☐
Cadmium	☐		☐	☐	☐
Chromium, total	☐		☐	☐	☐
Chromium, hexavalent	☐		☐	☐	☐
Cobalt	☐		☐	☐	☐
Copper	☐		☐	☐	☐
Iron	☐		☐	☐	☐
Lead	☐		☐	☐	☐
Manganese	☐		☐	☐	☐
Mercury	☐		☐	☐	☐
Mercury (Low Level)	☒		☒	☐	☐
Molybdenum	☐		☐	☐	☐
Nickel	☐		☐	☐	☐
Selenium	☐		☐	☐	☐
Silver	☐		☐	☐	☐
Thallium	☐		☐	☐	☐
Vanadium	☐		☐	☐	☐
Zinc	☐		☐	☐	☐
Demands					
5-day BOD	☐		☐	☐	☐
5-day Carbonaceous BOD	☐		☐	☐	☐
COD	☐		☐	☐	☐
TOC	☐		☐	☐	☐
Minerals					
Alkalinity, total (CaCO ₃)	☐		☐	☐	☐
Chloride	☐		☐	☐	☐
Fluoride	☐		☐	☐	☐
Hardness, total (CaCO ₃)	☐		☐	☐	☐
Specific conductance (25°C)	☐		☐	☐	☐
Sulfate	☐		☐	☐	☐
Total Dissolved Solids (180°C)	☐		☐	☐	☐
Nutrients					
Ammonia as N	☐		☐	☐	☐
Nitrate as N	☐		☐	☐	☐
Nitrite as N	☐		☐	☐	☐
Orthophosphate as P	☐		☐	☐	☐
Total Kjeldahl-Nitrogen as N	☐		☐	☐	☐
Total Phosphorus as P	☐		☐	☐	☐
Misc. Analytes					
Non-Filterable Residue (TSS)	☐		☐	☐	☐
Oil and Grease	☐		☐	☐	☐
pH	☐		☐	☐	☐
Total Cyanide	☐		☐	☐	☐
Total Phenolics (4-AAP)	☐		☐	☐	☐
Total Residual Chlorine	☐		☐	☐	☐
Total Residual Chlorine (Low Level)	☐		☐	☐	☐
Settleable Solids	☐		☐	☐	☐
Turbidity	☐		☐	☐	☐

Name Amy Vaal

Signature Amy Vaal

Date 8/11/16



A Waters Company

Amy Vail
Cadillac WWTP
200 N Lake Street
Cadillac, MI 49601
USA

DMR-QA 36 Final Report

DMR-QA Proficiency Testing

NPDES Permit: MI0020257

DMR-QA Study

Open Date: 03/18/16

Close Date: 07/01/16

Report Issued Date: 07/29/16



DMR-QA 36 Final Report

A Waters Company

NPDES Permit #: MI0020257
Permit Holder: Amy Vail

Lab Manager
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 07/29/16
Study Dates: 03/18/16 - 07/01/16

TNI Analyte Code	Analyte	Units	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Method Description	Study Mean	Study Standard Deviation	USEPA Lab Code	Study
DMRQA Hardness (cat# 580, lot# Q036-507)											
1960	Total Suspended Solids	mg/L	Acceptable	30.0	35.1	25.6 - 41.4	SM 2540 D-2011 2011	33.3	1.90	MI00082	DMRQA36
DMRQA pH (cat# 577, lot# Q036-977)											
1900	pH	S.U.	Acceptable	7.59	7.50	7.30 - 7.70	SM 4500-H+ B-2011 2011	7.54	0.0679	MI00082	DMRQA36
DMRQA Simple Nutrients (cat# 584, lot# Q036-505)											
1515	Ammonia as N	mg/L	Acceptable	10.5	11.4	9.10 - 13.6	SM 4500-NH3 D-2011 2011	11.4	0.832	MI00082	DMRQA36
DMRQA Complex Nutrients (cat# 579, lot# Q036-525)											
1910	Total phosphorus as P	mg/L	Acceptable	10.15	9.31	7.77 - 10.7	SM 4500-P E-2011 2011	9.41	0.447	MI00082	DMRQA36
DMRQA Demand (cat# 578, lot# Q036-516)											
1555	CBOD	mg/L	Acceptable	54.9	55.3	24.8 - 85.8	SM 5210 B-2011 2011	58.2	11.9	MI00082	DMRQA36



All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01

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Study #: DMR-QA 36

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A Waters Company

Study: DMR-QA 36
NPDES Permit #: MI0020257
Laboratory Name: Cadillac WWTP

Microbiology Results





DMR-QA 36 Final Report

A Waters Company

NPDES Permit #: MI0020257
Permit Holder: Amy Vail

Lab Manager
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 07/29/16
Study Dates: 03/18/16 - 07/01/16

TNI Analyte Code	Analyte	Units	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Method Description	Study Mean	Study Standard Deviation	USEPA Lab Code	Study
2530	Fecal Coliforms (MF)	CFU/100mL	Acceptable	278	1330	130 - 2690	SM 9222 D (m-FC)-2006 2006	592	427	MI00082	DMRQA36

DMRQA WasteWater™ Coliform MicrobE™ (cat# 576, lot# Q036-083)

Per Section 6.4.3 a of the 2009 TNI Standard, "The assigned values for quantitative microbiology analytes shall be equal to the mean of the assigned value verification and/or homogeneity testing per Sections 7.1 and 7.2". The final acceptance limits are derived from the calculated study mean and study standard deviation from laboratory-reported results. Disagreement between the assigned values and study means/acceptance limits are due to the inherent variability of microbiology methods and differences in the methods used by ERA and participant laboratories. For quantitative microbiology analytes, the assigned value is not used in the evaluation of laboratories.



All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01

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Study #: DMR-QA 36

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A Waters Company

Study: **DMR-QA 36**
NPDES Permit #: **MI0020257**
Laboratory Name: **Cadillac WWTP**

WET Results





A Waters Company

DMR-QA 36 Final Report

NPDES Permit #: MI0020257
Permit Holder: Amy Vail

Lab Manager
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 07/29/16
Study Dates: 03/18/16 - 07/01/16

Analyte Code	Test End Point	Performance Evaluation	Reported Value %	Assigned Value %	Acceptance Limits %	Method Description	Study Mean	Study Standard Deviation	USEPA Lab Code	Study
DMRWet Fathead minnow (Test Code 13) (cat# WET002, lot# Q036-002) Potassium chloride										
0754	LC50	Acceptable	59.4	58.4	31.7 - 85.2	EPA 2000.0 2002	58.4	13.4	MI01187	DMRWet36
DMRWet Fathead minnow (Test Code 15) (cat# WET004, lot# Q036-004) 7-day Short term Chronic, Daily Renewal, MHSF Potassium chloride										
0808	IC25 (ON) Growth	Acceptable	36.0	51.1	28.2 - 73.9	EPA 1000 2002	51.1	11.4	MI01187	DMRWet36
0810	NOEC (ON) Growth	Acceptable	25.0	50.0	25.0 - 100	EPA 1000 2002			MI01187	DMRWet36
0756	NOEC Survival	Acceptable	25.0	50.0	25.0 - 100	EPA 1000 2002			MI01187	DMRWet36
DMRWet Ceriodaphnia dubia (Test Code 19) (cat# WET008, lot# Q036-008) 48Hr., Acute Renewal, 25° C, MHSF Potassium chloride										
0764	LC50	Acceptable	37.3	42.6	19.1 - 66.2	EPA 2002	42.6	11.8	MI01187	DMRWet36
DMRWet Ceriodaphnia dubia (Test Code 21) (cat# WET010, lot# Q036-010) 7-day Short term Chronic, Daily Renewal, MHSF Potassium chloride										
0767	IC25 Reproduction	Acceptable	32.1	33.4	20.9 - 45.9	EPA 1002 2002	33.4	6.27	MI01187	DMRWet36
0768	NOEC Reproduction	Acceptable	25.0	25.0	12.5 - 50.0	EPA 1002 2002			MI01187	DMRWet36
0766	NOEC Survival	Acceptable	25.0	50.0	25.0 - 100	EPA 1002 2002			MI01187	DMRWet36



All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01

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Study #: DMR-QA 36

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CERTIFICATE OF RECOGNITION

ERA congratulates
CADILLAC WWTP

for your participation and successful evaluation in DMR-QA 36. We recognize the performance of this laboratory for achieving acceptable evaluations for the following analytes.

Ammonia as N	Fecal Coliforms (MIF)	Total phosphorus as P
CBOD	pH	Total Suspended Solids



Patrick Larson
Quality Officer

C454401



DMR-QA 36 Laboratory Exception Report

A Waters Company

William G. Rauch
President
Kar Labs
4425 Manchester Rd.
Kalamazoo, MI 49001
269-381-9666

EPA ID:
ERA Customer Number:
Report Issued:
Study Dates:

MI00041
K060501
07/29/16
03/18/16 - 07/01/16

2009 TNI Evaluation Checks

There are no values reported with < where the assigned value was greater than 0.

2009 TNI Not Acceptable Evaluations

There were no Not Acceptable evaluations for this study.



All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01
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Study # : DMR-QA 36



A Waters Company

Final Report Results For Laboratory Kar Labs





A Waters Company

2009 TNI Evaluation Report

Study: DMR-QA 36

ERA Customer Number: K060501

Laboratory Name: Kar Labs

Inorganic Results





DMR-QA 36 Final Report

A Waters Company

William G. Rauch
President
Kar Labs
4425 Manchester Rd.
Kalamazoo, MI 49001
269-381-9666

EPA ID:
ERA Customer Number:
Report Issued:
Study Dates:

MI00041
K060501
07/29/16
03/18/16 - 07/01/16

TNI Analyte Code	Analyte	Units	Reported Value	Assigned Value	Acceptance Limits	Performance Evaluation	Method Description	Analysis Date	Z Score	Study Mean	Study Standard Deviation	Analyst Name
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DMRQA Minerals (cat# 581, lot# Q036-506)

1505	Alkalinity as CaCO ₃	mg/L	45	42.0	35.7 - 48.3	Acceptable	SM2320B 21st ED 1997		1.32	41.7	2.53	
1575	Chloride	mg/L	124	124	109 - 139	Acceptable	EPA 300.0A		0.665	121	4.57	
1610	Conductivity at 25°C	µmhos/cm	606	605	545 - 666	Acceptable	EPA 120.1 1982		0.132	604	13.2	
1730	Fluoride	mg/L	3.2	3.39	2.74 - 3.90	Acceptable	SM4500F - C online		-0.792	3.32	0.151	
1125	Potassium	mg/L	35.2	38.5	30.8 - 46.2	Acceptable	EPA 200.7 4.4 1994		-1.54	37.1	1.26	
1155	Sodium	mg/L	90.9	98.9	79.1 - 119	Acceptable	EPA 200.7 4.4 1994		-0.875	94.3	3.85	
2000	Sulfate	mg/L	38	38.7	31.5 - 44.5	Acceptable	EPA 300.0A		0.215	37.5	2.26	
1955	Total Dissolved Solids at 180°C	mg/L		368	323 - 413	Not Reported				359	18.4	
1950	Total Solids at 105°C	mg/L		378	333 - 423	Not Reported				371	19.6	

DMRQA Hardness (cat# 580, lot# Q036-507)

1960	Total Suspended Solids	mg/L		35.1	25.6 - 41.4	Not Reported				33.3	1.90	
1035	Calcium	mg/L	94.8	94.0	79.9 - 108	Acceptable	EPA 200.7 4.4 1994		0.275	93.7	4.01	
1085	Magnesium	mg/L	14.2	14.7	12.5 - 16.9	Acceptable	EPA 200.7 4.4 1994		-0.605	14.6	0.615	
1550	Calcium Hardness as CaCO ₃	mg/L	237	235	200 - 270	Acceptable	SM2340B online		0.379	233	10.2	
1755	Total Hardness as CaCO ₃	mg/L	295	295	251 - 339	Acceptable	SM2340B online		0.223	293	9.71	

DMRQA Simple Nutrients (cat# 584, lot# Q036-505)

1515	Ammonia as N	mg/L	12.1	11.4	9.10 - 13.6	Acceptable	EPA 350.1 2 1993			11.4	0.832	
1820	Nitrate + Nitrite as N	mg/L	3.6	3.47	2.80 - 4.10	Acceptable	EPA 353.2 2 1993			3.49	0.196	
1810	Nitrate as N	mg/L	3.6	3.47	2.74 - 4.18	Acceptable	EPA 353.2 2 1993			3.48	0.229	
1870	ortho-Phosphate as P	mg/L	3.03	2.96	2.52 - 3.40	Acceptable	SM4500P E online			3.03	0.170	



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DMR-QA 36 Final Report

A Waters Company

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ERA Customer Number:
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Study Dates:

MI00041
K060501
07/29/16
03/18/16 - 07/01/16

TNI Analyte Code	Analyte	Units	Reported Value	Assigned Value	Acceptance Limits	Performance Evaluation	Method Description	Analysis Date	Z Score	Study Mean	Study Standard Deviation	Analyst Name
DMRQA Complex Nutrients (cat# 579, lot# Q036-525)												
1795	Total Kjeldahl Nitrogen	mg/L	24.9	24.5	18.4 - 29.6	Acceptable	EPA 351.2 2 1993		0.492	24.0	1.86	
1910	Total phosphorus as P	mg/L	9.39	9.31	7.77 - 10.7	Acceptable	SM4500-P B5 E		-0.0545	9.41	0.447	
DMRQA Demand (cat# 578, lot# Q036-516)												
1530	BOD	mg/L	71.6	61.1	32.1 - 90.0	Acceptable	SM5210B online			62.4	10.7	
1555	CBOD	mg/L	52.1	55.3	24.8 - 85.8	Acceptable	SM5210B online			58.2	11.9	
1565	COD	mg/L	95.3	99.0	75.2 - 119	Acceptable	EPA 410.4 2 1993			98.9	7.33	
2040	TOC	mg/L	40.3	39.1	32.4 - 45.6	Acceptable	SM5310C online			39.1	1.94	
DMRQA Oil & Grease Concentrate (cat# 4120, lot# Q036-4122)												
1860	n-Hexane Extractable Material(O&G)(Grav)	mg/L	146	151	111 - 172	Acceptable	EPA 1664		0.449	141	10.2	



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TNI Analyte Code	Analyte	Units	Reported Value	Assigned Value	Acceptance Limits	Performance Evaluation	Method Description	Analysis Date	Z Score	Study Mean	Study Standard Deviation	Analyst Name
DMRQA Trace Metals (cat# 586, lot# Q036-500)												
1000	Aluminum	µg/L	1759	1770	1460 - 2030	Acceptable	EPA 200.7 & 200.8	5/17/2016	-0.258	1790	113	
1005	Antimony	µg/L	457	496	400 - 577	Acceptable	EPA 200.7 & 200.8	5/17/2016	-1.27	486	22.8	
1010	Arsenic	µg/L	695	751	635 - 857	Acceptable	EPA 200.7 & 200.8	5/17/2016	-1.55	739	28.6	
1015	Barium	µg/L	1585	1620	1380 - 1860	Acceptable	EPA 200.7 & 200.8	5/17/2016	-0.637	1620	53.8	
1020	Beryllium	µg/L	303	302	257 - 347	Acceptable	EPA 200.7 & 200.8	5/17/2016	0.106	302	12.0	
1025	Boron	µg/L	1060	1030	876 - 1180	Acceptable	EPA 200.7 & 200.8	5/17/2016	0.420	1040	50.1	
1030	Cadmium	µg/L	368	385	327 - 443	Acceptable	EPA 200.7 & 200.8	5/17/2016	-0.815	381	15.5	
1040	Chromium	µg/L	235	251	213 - 289	Acceptable	EPA 200.7 & 200.8	5/17/2016	-1.69	253	10.7	
1050	Cobalt	µg/L	480	487	414 - 560	Acceptable	EPA 200.7 & 200.8	5/17/2016	-1.26	507	21.1	
1055	Copper	µg/L	303	323	275 - 371	Acceptable	EPA 200.7 & 200.8	5/17/2016	-1.47	324	14.1	
1070	Iron	µg/L	1730	1780	1510 - 2050	Acceptable	EPA 200.7 & 200.8	5/17/2016	-0.691	1780	76.5	
1075	Lead	µg/L	718	731	621 - 841	Acceptable	EPA 200.7 & 200.8	5/17/2016	-0.459	731	28.8	
1090	Manganese	µg/L	1631	1680	1430 - 1930	Acceptable	EPA 200.7 & 200.8	5/17/2016	-0.973	1700	70.8	
1100	Molybdenum	µg/L	397	414	358 - 466	Acceptable	EPA 200.7 & 200.8	5/17/2016	-0.701	408	15.6	
1105	Nickel	µg/L	579	603	527 - 683	Acceptable	EPA 200.7 & 200.8	5/17/2016	-1.65	609	18.3	
1140	Selenium	µg/L	245	248	211 - 285	Acceptable	EPA 200.7 & 200.8	5/17/2016	0.246	242	11.1	
1150	Silver	µg/L	540	563	479 - 647	Acceptable	EPA 200.7 & 200.8	5/17/2016	-0.595	557	27.9	
1160	Strontium	µg/L	259	267	227 - 307	Acceptable	EPA 200.7 & 200.8	5/17/2016	-1.01	271	12.1	
1165	Thallium	µg/L	260	261	208 - 308	Acceptable	EPA 200.7 & 200.8	5/17/2016	0.0763	259	12.4	
1185	Vanadium	µg/L	511	543	462 - 624	Acceptable	EPA 200.7 & 200.8	5/17/2016	-1.71	539	16.7	



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DMR-QA 36 Final Report

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TNI Analyte Code	Analyte	Units	Reported Value	Assigned Value	Acceptance Limits	Performance Evaluation	Method Description	Analysis Date	Z Score	Study Mean	Study Standard Deviation	Analyst Name
DMRQA Trace Metals (cat# 586, lot# Q036-500) (Continued)												
1190	Zinc	µg/L		422	359 - 485	Not Reported				422	19.8	



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DMRQA Trace Metals (cat# 586, lot# Q036-500)												
1000	Aluminum	µg/L		1770	1460 - 2030	Not Reported				1790	113	
1005	Antimony	µg/L		496	400 - 577	Not Reported				486	22.8	
1010	Arsenic	µg/L		751	635 - 857	Not Reported				739	28.6	
1015	Barium	µg/L		1620	1380 - 1860	Not Reported				1620	53.8	
1020	Beryllium	µg/L		302	257 - 347	Not Reported				302	12.0	
1025	Boron	µg/L		1030	876 - 1180	Not Reported				1040	50.1	
1030	Cadmium	µg/L		385	327 - 443	Not Reported				381	15.5	
1040	Chromium	µg/L		251	213 - 289	Not Reported				253	10.7	
1050	Cobalt	µg/L		487	414 - 560	Not Reported				507	21.1	
1055	Copper	µg/L		323	275 - 371	Not Reported				324	14.1	
1070	Iron	µg/L		1780	1510 - 2050	Not Reported				1780	76.5	
1075	Lead	µg/L		731	621 - 841	Not Reported				731	28.8	
1090	Manganese	µg/L		1680	1430 - 1930	Not Reported				1700	70.8	
1100	Molybdenum	µg/L		414	358 - 466	Not Reported				408	15.6	
1105	Nickel	µg/L		603	527 - 683	Not Reported				609	18.3	
1140	Selenium	µg/L		248	211 - 285	Not Reported				242	11.1	
1150	Silver	µg/L		563	479 - 647	Not Reported				557	27.9	
1160	Strontium	µg/L		267	227 - 307	Not Reported				271	12.1	
1165	Thallium	µg/L		261	208 - 308	Not Reported				259	12.4	
1185	Vanadium	µg/L		543	462 - 624	Not Reported				539	16.7	



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TNI Analyte Code	Analyte	Units	Reported Value	Assigned Value	Acceptance Limits	Performance Evaluation	Method Description	Analysis Date	Z Score	Study Mean	Study Standard Deviation	Analyst Name
DMRQA Trace Metals (cat# 586, lot# Q036-500) (Continued)												
1190	Zinc	µg/L	412	422	359 - 485	Acceptable	EPA 200.8.5.4 1994	5/17/2016	-0.514	422	19.8	
DMRQA Low-Level Mercury (cat# 896, lot# Q036-931)												
1095	Low Level Mercury	ng/L	32.1	32.1	20.1 - 43.9	Acceptable	EPA 1631E 2002		0.498	31.2	1.86	
DMRQA Total Cyanide (cat# 588, lot# Q036-502)												
1645	Cyanide, total	mg/L	0.5992	0.591	0.384 - 0.798	Acceptable	SM4500CN C E		-0.663	0.638	0.0590	
1510	Amenable Cyanide	mg/L		0.205	0.133 - 0.277	Not Reported				0.272	0.0820	
DMRQA Total Phenolics (4-AAP) (cat# 589, lot# Q036-515)												
1905	Phenolics, total	mg/L	0.815	0.640	0.317 - 0.964	Acceptable	EPA 420.4 1 1993		0.874	0.714	0.115	
DMRQA Total Residual Chlorine (cat# 587, lot# Q036-501)												
1940	Total Residual Chlorine	mg/L	1.606	1.40	1.04 - 1.66	Acceptable	SM4500Cl online		2.08	1.38	0.107	



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A Waters Company

Study: DMR-QA 36
NPDES Permit #: MI0020257
Laboratory Name: Cadillac WWTP

Inorganic Results





A Waters Company

DMR-QA 36 Final Report

NPDES Permit #: M10020257
Permit Holder: Amy Vail
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ERA Customer Number: C454401
Report Issued: 07/29/16
Study Dates: 03/18/16 - 07/01/16

TNI Analyte Code	Analyte	Units	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Method Description	Study Mean	Study Standard Deviation	USEPA Lab Code	Study
DMRQA Hardness (cat# 580, lot# Q036-507)											
1960	Total Suspended Solids	mg/L	Acceptable	30.0	35.1	25.6 - 41.4	SM 2540 D-2011 2011	33.3	1.90	MI00082	DMRQA36
DMRQA pH (cat# 577, lot# Q036-977)											
1900	pH	S.U.	Acceptable	7.59	7.50	7.30 - 7.70	SM 4500-H+ B-2011 2011	7.54	0.0679	MI00082	DMRQA36
DMRQA Simple Nutrients (cat# 584, lot# Q036-505)											
1515	Ammonia as N	mg/L	Acceptable	10.5	11.4	9.10 - 13.6	SM 4500-NH3 D-2011 2011	11.4	0.832	MI00082	DMRQA36
DMRQA Complex Nutrients (cat# 579, lot# Q036-525)											
1910	Total phosphorus as P	mg/L	Acceptable	10.15	9.31	7.77 - 10.7	SM 4500-P E-2011 2011	9.41	0.447	MI00082	DMRQA36
DMRQA Demand (cat# 578, lot# Q036-516)											
1555	CBOD	mg/L	Acceptable	54.9	56.3	24.8 - 85.8	SM 5210 B-2011 2011	58.2	11.9	MI00082	DMRQA36



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Study #: DMR-QA 36

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A Waters Company

Study: DMR-QA 36

NPDES Permit #: MI0020257

Laboratory Name: Cadillac WWTP

Microbiology Results



Study #: DMR-QA 36

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DMR-QA 36 Final Report

A Waters Company

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ERA Customer Number: C454401
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Study Dates: 03/18/16 - 07/01/16

TNI Analyte Code	Analyte	Units	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Method Description	Study Mean	Study Standard Deviation	USEPA Lab Code	Study
DMRQA WasteWater™ Coliform MicrobE™ (cat# 576, lot# Q036-083)											
2530	Fecal Coliforms (MF)	CFU/100mL	Acceptable	278	1330	130 - 2690	SM 9222 D (m-FC)-2006	592	427	MI00082	DMRQA36

Per Section 6.4.3 a of the 2009 TNI Standard, "The assigned values for quantitative microbiology analytes shall be equal to the mean of the assigned value verification and/or homogeneity testing per Sections 7.1 and 7.2". The final acceptance limits are derived from the calculated study mean and study standard deviation from laboratory-reported results. Disagreement between the assigned values and study means/acceptance limits are due to the inherent variability of microbiology methods and differences in the methods used by ERA and participant laboratories. For quantitative microbiology analytes, the assigned value is not used in the evaluation of laboratories.



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Study: DMR-QA 36

NPDES Permit #: MI0020257

Laboratory Name: Cadillac WWTP

WET Results



Study #: DMR-QA 36

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Analyte Code	Test End Point	Performance Evaluation	Reported Value %	Assigned Value %	Acceptance Limits %	Method Description	Study Mean	Study Standard Deviation	USEPA Lab Code	Study
DMRWet Fathead minnow (Test Code 13) (cat# WET002, lot# Q036-002)										
0754	LC50	Acceptable	59.4	58.4	31.7 - 85.2	EPA 2000.0 2002	58.4	13.4	MI01187	DMRWet36
DMRWet Fathead minnow (Test Code 15) (cat# WET004, lot# Q036-004)										
7-day Short term Chronic, Daily Renewal, MHSF Potassium chloride										
0808	IC25 (ON) Growth	Acceptable	36.0	51.1	28.2 - 73.9	EPA 1000 2002	51.1	11.4	MI01187	DMRWet36
0810	NOEC (ON) Growth	Acceptable	25.0	50.0	25.0 - 100	EPA 1000 2002			MI01187	DMRWet36
0756	NOEC Survival	Acceptable	25.0	50.0	25.0 - 100	EPA 1000 2002			MI01187	DMRWet36
DMRWet Ceriodaphnia dubia (Test Code 19) (cat# WET008, lot# Q036-008)										
48Hr., Acute Renewal, 25° C, MHSF Potassium chloride										
0764	LC50	Acceptable	37.3	42.6	19.1 - 66.2	EPA 2002	42.6	11.8	MI01187	DMRWet36
DMRWet Ceriodaphnia dubia (Test Code 21) (cat# WET010, lot# Q036-010)										
7-day Short term Chronic, Daily Renewal, MHSF Potassium chloride										
0767	IC25 Reproduction	Acceptable	32.1	33.4	20.9 - 45.9	EPA 1002 2002	33.4	6.27	MI01187	DMRWet36
0768	NOEC Reproduction	Acceptable	25.0	25.0	12.5 - 50.0	EPA 1002 2002			MI01187	DMRWet36
0766	NOEC Survival	Acceptable	25.0	50.0	25.0 - 100	EPA 1002 2002			MI01187	DMRWet36



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